Role Prompt

You are a reward engineer trying to write reward functions to solve reinforcement learning tasks as effective as possible.

Platoon Environment Code

```python
class Platoon_Env():
    def __init__(self,
                 num_vehicle=100,
                 points=None,
                 start_time=None):

        self.num_vehicle = num_vehicle
        # initial start time and path
        self.set_initial_scenarios()

        # initial map adjacency_matrix
        self.DG, self.adjacency_matrix, self.edges = self.get_topo()

        # initial state
        self.reset()

    def step(self, action):
        for c_edge in cur_edge_dict:
            cur_edge_dict[c_edge].sort(key=lambda x: -x[1])
            # firstly, update platoon state
            for idx, car in enumerate(cur_edge_dict[c_edge]):
                car_idx = car[0]
                distance = car[1]
                velocity = self.speed_list[action[car_idx]]
                if action[car_idx] == 1:
                    overlap_num = 0
```

Platoon Coordination Task Description

Minimize delay time as much as possible while **forming more platoons**.

Large Language Model

Role

Env Code

&

Task

Analysis Results

&

Reward Codes

Feedback & Evolution

New Reward Codes

Code Analysis Results

1. Implementation Details
(1) The code uses the following packages: ...

2. Environment Structure
(1) Observation Space: ...
(2) Action Space: ...
(3) State Space: ...

3. Agent Design and Interaction Dynamics
(1) The agents are homogeneous.
...
...

Initial Rewards

Evolutionary Update

Reward Function Pool

Reward Function Code 1

```python
def calculate_reward(args):
```

Reward Function Code 2

```python
def calculate_reward(args):
    reward = 0
    for car_idx in range(num_vehicle):
        reward_fuel_i, reward_delay_i, reward_platoon_i = 0, 0, 0
        # fuel reward
        reward_fuel_i = ...
        # time reward
        reward_delay_i = ...
        # platoon reward
        reward_platoon_i = ...

        reward += w1 * reward_fuel_i + w2 * reward_delay_i + \
                  w3 * reward_platoon_i

    return reward
```

.....

Reward Function Code n

```python
def calculate_reward(args):
    reward = 0
    for car_idx in range(num_vehicle):
        reward_fuel_i, reward_delay_i = 0, 0
        # fuel reward
        reward_fuel_i = ...
        # time reward
        reward_delay_i = ...

        reward += w1 * reward_fuel_i + w2 * reward_delay_i

    return reward
```

Directed Evolution Module

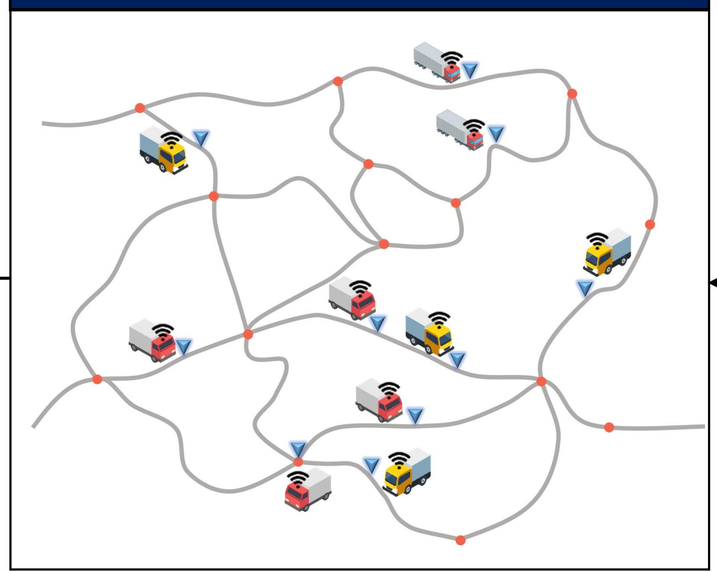
- Modify Prompt 1
- Modify Prompt 2
- Modify Prompt 3
- Create Prompt 1

Best Reward Function Code

Best Reward Training Feedback

Evaluate & Rank Training Results

Platoon Coordination Environment



Parallel Training

Initial Inference

Loop Evolution